

DATA ARTICLE

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected with species distribution modelling using standardized data layers representing physical, chemical and biological conditions of the global ocean. Yet, the available data layers (1) have not been updated to incorporate data of the Sixth Phase of the Coupled Model Intercomparison Project (CMIP6), which comprise the Shared Socioeconomic Pathway (SSP) scenarios; (2) consider a limited number of Earth System Models (ESMs), and (3) miss important variables expected to influence future biodiversity distributions. These limitations might undermine biodiversity impact assessments, by failing to integrate them within the context of the most up-to-date climate change projections, raising the uncertainty in estimates and misinterpreting the exposure of biodiversity to extreme conditions. Here, we provide a significant update of Bio-ORACLE, extending biologically relevant data layers from present-day conditions to the end of the 21st century Shared Socioeconomic Pathway scenarios based on a multi-model ensemble with data from CMIP6. Alongside, we provide R and Python packages for seamless integration in modelling workflows. The data layers aim to enhance the understanding of the potential impacts of climate change on biodiversity and to support well-informed research, conservation and management.

Main Types of Variable Contained: Surface and benthic layers for, chlorophyll-*a*, diffuse attenuation coefficient, dissolved iron, dissolved oxygen, nitrate, ocean temperature, pH, phosphate, photosynthetic active radiation, total phytoplankton, total cloud fraction, salinity, silicate, sea-water direction, sea-water velocity, topographic slope, topographic aspect, terrain ruggedness index, topographic position index and

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bathymetry, and surface layers for air temperature, mixed layer depth, sea-ice cover and sea-ice thickness.

Spatial Location and Grain: Global at 0.05° resolution.

Time Period and Grain: Decadal from present-day to the end of the 21st century (2000–2100).

Major Taxa and Level of Measurement: Marine biodiversity associated with surface and epibenthic habitats.

Software Format: A package of functions developed for Python and R software.

KEYWORDS

Bio-ORACLE, climate change projections, environmental data, macroecology, shared socioeconomic pathway (SSP) scenarios, species distribution modelling

1 | INTRODUCTION

Research quantifying the impacts of climate change on biodiversity relies heavily on species distribution modelling (Peterson et al., 2011) using high-resolution data layers representing physical, chemical and biological conditions of the global ocean (Assis, Tyberghein, et al., 2017; Fick & Hijmans, 2017; Tyberghein et al., 2012). Initiatives providing essential climate layers uniform in extent and resolution, like MERRAclim (Vega et al., 2017) and WorldClim (Fick & Hijmans, 2017) for terrestrial environments and the World Ocean Atlas (Levitus et al., 2013), Hexacoral (Fautin & Buddemeier, 2006), Marspec (Sbrocco & Barber, 2013), Bio-ORACLE (Assis, Tyberghein, et al., 2017; Tyberghein et al., 2012) and OCLE (de la Hoz et al., 2018) for marine environments, established solid standards to estimate future climate-induced pressures on the world's ecosystems and associated ecosystem services (Assis, Bercibar, et al., 2017; Assis, Serrão, et al., 2022; Gouvêa et al., 2020, 2022; Martins et al., 2021). But despite their relevance, current marine data sets have three main limitations that can preclude the development of well-informed research, conservation and management strategies (Arafah-Dalau et al., 2021; Arneth et al., 2020).

First, current standardized climate layers have not been updated to incorporate data of the Sixth Phase of the Coupled Model Intercomparison Project (CMIP6), which serves as the core contribution to the latest Intergovernmental Panel on Climate Change Assessment Report (IPCC). The CMIP6 dataset includes an updated set of variables (Séférian et al., 2020), which better capture the spatial and temporal variability of physical and biogeochemical properties of the global ocean (Séférian et al., 2020), and the Shared Socioeconomic Pathway (SSP) scenarios, a set of narratives on possible global change trajectories throughout the 21st century, with and without climate policy responses (Riahi et al., 2017; van Vuuren et al., 2017). Compared to the previous Representative Concentration Pathway scenarios (van Vuuren et al., 2011), despite producing similar forcing pathways, the SSPs use more up-to-date socioeconomic assumptions (Burgess et al., 2023) and encompass the Paris Agreement expectations, which allows estimating the

biodiversity benefits of compliance to international climate policies (Martins et al., 2021; Sanderson et al., 2016).

Second, current standardized climate layers projecting future conditions consider a limited number of Earth System Models (ESMs) (Schoeman et al., 2023). These are intensive computer simulations tuned by specific parameters and conditions (e.g. greenhouse gas concentrations, solar forcing, land-use, as well as atmospheric and ocean dynamics (Mechoso & Arakawa, 2015)) that can resolve global atmospheric and oceanic conditions across different time scales. There are, however, inherent uncertainties of ESMs projections due to the complexity of the Earth's climate system, as well as the limited understanding and the challenges in incorporating particular processes in the models. Accordingly, the multi-model ensemble offers a straightforward approach in species distribution modelling (Araújo & New, 2007; Assis, Tyberghein, et al., 2017; Gouvêa et al., 2022), yet at the cost of incorporating ESM that do not perform well in reproducing historical climates over time (please refer to the 'hot model' problem; Hausfather et al., 2022). The previous version of Bio-ORACLE (Assis, Tyberghein, et al., 2017) was based on a multi-model ensemble, yet limited to two ESMs from the dozens available, limiting the breadth of projections of climate change impact.

Third, current standardized climate layers miss projections of biologically meaningful variables beyond temperature, salinity and sea ice conditions (Assis, Tyberghein, et al., 2017; Tyberghein et al., 2012). For instance, dissolved oxygen, primary productivity and pH (among other variables) are expected to change in the coming decades and are emerging as limiting factors for marine biodiversity (Fragkopoulou et al., 2021; Krumhardt et al., 2017; Martins et al., 2021; Shi et al., 2022). Failing to consider them in climate change projections might misinterpret biodiversity exposure to novel detrimental conditions.

To address these gaps in the marine climate data space, we provide a substantial extension of Bio-ORACLE, including a set of essential variables for bioclimatic modelling from present-day conditions to the end of the 21st century SSP scenarios, for both surface and benthic (i.e. along the seafloor) conditions. These were built by ensembling numerous ESMs provided by the CMIP6, from the higher (SSP1, in line with the Paris Agreement) to the lower

mitigation (SSP5) trajectories of global change. These new layers are provided under the FAIR principle (Findability, Accessibility, Interoperability and Reusability (Wilkinson et al., 2016)) with a finer spatial and temporal resolution (0.05° spatial resolution; decadal climatologies). Alongside the data layers, R and Python packages were developed for programmatic data access and easy integration in available bioclimatic modelling workflows (Naimi & Araújo, 2016; Thuiller et al., 2009). The extension of Bio-ORACLE aims to improve understanding of the potential impacts of climate change on marine organisms and support well-informed research, conservation and management.

2 | MARINE DATA LAYERS

The current data descriptor provides information on the approach used to generate a set of essential variables for bioclimatic modelling. This comprises the acquisition of physical, chemical, biological and topographic data and the techniques used to generate variables for present-day conditions and the SSP scenarios of future climate change.

Data for present-day conditions (period 2000 to 2020) were acquired at a monthly basis from the Global Ocean Physics Reanalysis and Forecast and the Global Ocean Biogeochemistry Analysis and Forecast, two pre-processed re-analyses of Copernicus Marine Environment Monitoring Service (<https://data.marine.copernicus.eu/products>), provided at a 0.08° and 0.25° degree resolution with 50 and 75 vertical levels, respectively. These integrate and assimilate a comprehensive range of satellite and in situ data (Jean-Michel et al., 2021). Specifically for Photosynthetic Active Radiation and Diffuse Attenuation Coefficient, data were acquired as monthly averages from GlobColour (<https://www.globcolour.info>), a data set at a 0.04° resolution that merges MERIS, MODIS and SeaWiFS sensors (Maritorena et al., 2010). Data for the SSP1-1.9, SSP1-2.6, SSP2-4.6, SSP3-7.0 and SSP5.8.5 scenarios (period 2000 to 2100) were acquired at the monthly basis from numerous ESMs provided by the CMIP phase 6 (Earth System Grid Federation), namely, ACCESS-ESM1-5 (Australian Community Climate and Earth System Simulator; 1.875 × 1.25° resolution and 38 vertical levels), CanESM5 (Canadian Earth System; 1° resolution and 45 vertical levels), CESM2-WACCM (Community Earth System Model 2—Whole Atmosphere Community Climate Model; ~0.88 × 0.56° resolution and 60 vertical levels), CNRM-ESM2-1 (Centre National de Recherches Météorologiques; 0.25° resolution and 75 vertical levels), GFDL-ESM4 (Geophysical Fluid Dynamics Laboratory; 0.25° resolution and 75 vertical levels), GISS-E2-1-G (Goddard Institute for Space Studies; 1 × 1.25° resolution and 40 vertical levels), IPSL-CM6A-LR (Institut Pierre Simon Laplace; 1° resolution and 75 vertical levels), MIROC-ES2L (Model for Interdisciplinary Research on Climate; 1° resolution and 63 vertical levels), MPI-ESM1-2-LR (Max Plank Institute; 1.5° resolution and 40 vertical levels), MRI-ESM2-0 (Meteorological Research Institute; 1° resolution and 63 vertical levels) and UKESM1-0-LL (United Kingdom Earth System Model; 1° resolution and 75 vertical levels).

The monthly data of present-day and future conditions were used to generate six statistics per decade and variable: the average, maximum and minimum records of a given decade, long-term average of the yearly maxima and minima of a given decade (e.g. the average temperature of the warmest month in the period 2000–2010), and range, which represents the average absolute difference between the maximum and minimum records per year. These statistics were produced for air temperature, chlorophyll-a, diffuse attenuation coefficient, dissolved iron, dissolved oxygen, mixed layer depth, nitrate, ocean temperature, pH, phosphate, photosynthetically active radiation, primary productivity (total phytoplankton), total cloud fraction, salinity, silicate, sea-ice cover, sea-ice thickness and sea-water direction and velocity (Table 1).

To produce gridded layers with uniform spatial extent and resolution, the statistics generated for present-day conditions were interpolated to 0.05° (approx. 5.5 km at the equator) with inverse distance weighting (IDW). This is a well-established algorithm in climate change research (Assis, Tyberghein, et al., 2017; Ozelkan et al., 2016) and was chosen based on a performance and tuning test against the Kriging algorithm (Supplement S1). The inverse distance weighting algorithm fitted the 8 nearest values of each focal cell with an inverse distance power of 2. For surface layers, the algorithm performed bilinear interpolation, considering the position of each cell (i.e. longitude and latitude) and the information comprised in the top vertical levels (i.e. defining the surface of the ocean) of each data set (e.g. Copernicus) and variable. For benthic layers, the algorithm performed trilinear interpolation, by further considering the depth of each cell and the multiple vertical levels of each data set and variable (Assis, Tyberghein, et al., 2017; Boavida et al., 2016). To this end, the depth of each cell was extracted from the general bathymetric chart of the oceans (GEBCO_2023 Grid), a global terrain model providing elevation data at a 0.004° resolution (GEBCO Bathymetric Compilation Group 2023, 2023). Because focal cells at 0.05° comprise a wide range of depth values, the benthic layers were developed for the minimum, average and maximum depths within focal cells, as in Bio-ORACLE version 2.0 (Assis, Tyberghein, et al., 2017). The process of interpolation does not add local detail in climate data, but rather smooths it across the generated layers.

The layers projecting future conditions were produced with the change-factor method (also known as the delta change method). This involved adding the changes projected in climate by the ESMs (as interpolated anomalies, i.e., differences) to high-resolution climatologies representing present-day conditions (Assis, Tyberghein, et al., 2017; Maraun et al., 2017; Schoeman et al., 2023). In detail, the method was based on (1) computing the climate change anomaly (i.e. difference) between the future and the historical conditions with data of each ESM, at their native spatial resolution, and for the period of 2000–2014 (ESM projections start in the year 2015); (2) interpolating the climate change anomalies to the common 0.05° resolution with inverse distance weighting to be uniform in extent and resolution, as described for the present-day statistics; (3) averaging the interpolated climate change anomalies of the multiple ESM; and (4) applying the averaged downscaled anomalies to the

TABLE 1 Marine data layers and units, realm (surface layers, benthic layers or both, where appropriate), accuracy assessed with in situ quality-control data (for the variables comprised in GLODAP (Lauvset et al., 2021; Olsen et al., 2016); MAE: mean absolute error; RMSE: root mean square error; Cor: Pearson's correlation; additional estimates per depth in Supplement S3), number of quality-control records (*n*), range of values for present conditions and change (anomaly) between the present (decade 2010–2020) and the future (decade 2090–2100) under contrasting scenarios SSP1-1.9 and SSP5-8.5.

Layer	Surface	Benthic	MAE	RMSE	Cor	<i>n</i>	Range	Future change (SSP1-1.9; SSP5-8.5)
Air temperature (°C)	X		-	-	-	-	-57.96; 40.97	0.72 ± 1.08; 4.67 ± 3.53
Chlorophyll- <i>a</i> concentration (mmol m ⁻³)	X	X	-	-	-	-	0; 21.95	0 ± 0.03; 0.01 ± 0.03
Diffuse attenuation coefficient (m ⁻¹)	X		-	-	-	-	0; 1.19	-
Dissolved iron concentration (mmol m ⁻³)	X	X	-	-	-	-	0; 0.101	-0.01 ± 0.01; -0.01 ± 0.01
Dissolved oxygen concentration (mmol m ⁻³)	X	X	5.222	25.648	0.930	558,720	41.70; 456.98	-1.26 ± 4.93; -14.33 ± 11.33
Mixed layer depth (m)	X		-	-	-	-	0; 3728.89	0.27 ± 11.36; -7.19 ± 19.35
Nitrate concentration (mmol m ⁻³)	X	X	0.467	3.082	0.946	642,133	0; 303.75	-0.38 ± 0.91; -1.11 ± 1.29
Ocean temperature (°C)	X	X	0.141	0.823	0.991	823,531	-1.94; 36.49	0.38 ± 0.40; 2.97 ± 0.98
pH	X	X	-	-	-	-	7; 8.53	-0.03 ± 0.02; -0.39 ± 0.04
Phosphate concentration (mmol m ⁻³)	X	X	0.028	0.218	0.948	612,203	0; 4.38	-0.03 ± 0.09; -0.10 ± 0.11
Photo. active radiation (E m ⁻² yr ⁻³)	X		-	-	-	-	0; 69.13	-
Total phytoplankton concentration (mmol m ⁻³)	X	X	-	-	-	-	0.02; 57.20	-0.01 ± 0.16; -0.10 ± 0.29
Total cloud fraction (fraction)	X		-	-	-	-	0; 1	0.01 ± 0.04; -0.01 ± 0.05
Salinity	X	X	0.012	0.217	0.974	824,173	0; 47.54	-0.01 ± 0.32; -0.22 ± 0.51
Silicate concentration (mmol m ⁻³)	X	X	1.762	9.484	0.929	645,495	0.23; 680.78	-0.83 ± 3.33; -1.60 ± 3.53
Sea-ice cover (fraction)	X		-	-	-	-	0; 1	-0.02 ± 0.05; -0.09 ± 0.20
Sea-ice thickness (m)	X		-	-	-	-	0; 7.87	-0.13 ± 0.37; -0.28 ± 0.68
Sea-water direction (°)	X	X	-	-	-	-	0; 360	0.45 ± 40.66; 2.59 ± 45.39
Sea-water velocity (m/s)	X	X	-	-	-	-	0; 3.05	0.02 ± 0.05; 0.02 ± 0.05
Topographic slope		X	-	-	-	-	0; 30.71	-
Topographic aspect		X	-	-	-	-	0; 360	-
Terrain ruggedness index		X	-	-	-	-	0; 2512.92	-
Topographic position index		X	-	-	-	-	-1963; 2512	-
Minimum depth (m)		X	-	-	-	-	-10,363; 0	-
Average depth (m)		X	-	-	-	-	-10,699; 0	-
Maximum depth (m)		X	-	-	-	-	-10,977; 0	-

downscaled layers defining the conditions of the present-day period. This approach, considering the magnitude of climate change (i.e. the change-factor), allowed removing mean state biases of the ESM (Schoeman et al., 2023), while providing a high confidence level in climate change projections (Hall & Hall, 2014), linking greenhouse gas emissions with global climate change with and without climate policy responses (Riahi et al., 2017). The change-factor approach was not used in diffuse attenuation and photosynthetically active radiation, as these variables are not provided by the CMIP. The standard deviation of the ESM projected data was also computed (Supplement S2).

Additional topographic layers, namely, slope (expressing depth changes over distance), aspect (expressing the direction that slope faces), terrain ruggedness index (expressing depth changes between adjacent focal cells), topographic position index (comparing the depth of focal cells to the mean elevation of adjacent focal cells), as well as the minimum, average and maximum depth of each focal cell, were generated at 0.05° resolution based on the general bathymetric chart of the oceans (GEBCO Bathymetric Compilation Group 2023, 2023).

All layers were archived as NetCDF (network Common Data Form) and deposited into an ERDDAP (Environmental Research Division's Data Access Program) server to facilitate filtering and downloading of the layers in common data formats (Wilson et al., 2020). Additionally, Python (pyo_oracle) and R (biooracle) packages were developed for facilitated data retrieval and improved integration in available frameworks of bioclimatic modelling (Naimi & Araújo, 2016; Thuiller et al., 2009). Such packages act as clients for ERDDAP's REST API, an interoperable web protocol for data transfer, and thus can be used for integration into most generic web-based applications (Table 2).

TABLE 2 List of R and Python functions to facilitate listing and extraction of data layers (refer to the permanent GitHub repository for additional information; <https://github.com/bio-oracle>).

Function	Language	Description
<code>list_layers()</code>	R and Python	Lists the data layers available in the Bio-ORACLE ERDDAP server, either as a list or a data frame containing metadata. Users may subset layers based on attributes, such as variable or SSP scenario
<code>download_layers()</code>	R and Python	Downloads one or more data layers from the Bio-ORACLE ERDDAP server. Users may filter data using attributes, such as sets of coordinates or time periods, and select the data format to be downloaded*
<code>list_local_data()</code>	R and Python	Lists local data that have been downloaded by the Bio-ORACLE client
<code>config</code>	R and Python	Shows configuration values, that is, the path to the local data directory and the address of the Bio-ORACLE server

*Refer to the ERDDAP server for the complete list of data formats (<https://erddap.bio-oracle.org/erddap/griddap/documentation.html#fileType>).

3 | DATA RECORDS

The data set (Assis, 2023) comprises: (1) decadal time series of six statistics for each of 19 essential physical, chemical and biological variables at the global scale, for surface and benthic conditions, at a spatial resolution of 0.05°, and a temporal resolution of 10 decadal steps, from 2000 to 2100, under the scenarios SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP4-6.0 and SSP5-8.5; (2) seven topographic layers at a spatial resolution of 0.05 (Table 1).

4 | TECHNICAL VALIDATION

The reliability of data layers was estimated by means of cross-validation against quality-controlled data provided by the Global Ocean Data Analysis Project (GLODAP [Lauvset et al., 2021; Olsen et al., 2016]), for the variables available in this data set (dissolved oxygen, nitrate, phosphate, salinity, silicate and temperature). Following additional applications (Assis et al., 2015; Assis, Neiva, et al., 2022; Assis, Tyberghein, et al., 2017; Boavida et al., 2016; Davies & Guinotte, 2011), GLODAP records, which are in situ observations, were compared at every given location with the data used to develop the layers. To this end, the data was interpolated with inverse distance weighting, as previously described, to the locations reported on GLODAP (i.e. longitude, latitude and depth). Then, the paired records were statistically analysed with Pearson's correlation coefficient, root mean square error and mean absolute error. The difference between paired records was also mapped globally onto a 2.5 degrees grid (average of differences within grid cells [Assis, Tyberghein, et al., 2017; Davies & Guinotte, 2011]).

Overall, the layers mirrored the climatic patterns of quality-controlled data, as verified elsewhere (Assis, Neiva, et al., 2022; Assis, Tyberghein, et al., 2017). All variables retrieved high correlation coefficients, ranging from 0.93 to 0.99, and low error rates (Supplement S3). Nitrate, phosphate, salinity and temperature displayed discrete anomalies restricted to specific regions (e.g. the transition between the warm and cold temperate Northwest Atlantic and Pacific oceans), with no relationship with depth. The errors for dissolved molecular oxygen and silicate were mostly distributed in the Southern Ocean, also with no relationship with depth (Supplement S3).

The marine data layers produced with information from the Global Ocean Physics Reanalysis and Forecast and the Global Ocean Biogeochemistry Analysis and Forecast from the Copernicus Marine Environment Monitoring Service, and the numerous ESMs from CMIP phase 6, have climate data in all gridded cells. However, 7.81% of the cells of Photosynthetic Active Radiation and 4.22% of the cells of Diffuse Attenuation Coefficient have missing information at latitudes above 70° N/S. This is due to missing satellite information from MERIS, MODIS and SeaWiFS sensors (Maritorena et al., 2010) at such higher latitudes, as reported elsewhere (Tyberghein et al., 2012).

5 | USAGE NOTES

This major update of Bio-ORACLE presents twenty-six physical, chemical, biological and topographic marine data layers, with global coverage and uniform grid system. These layers, provided with improved 0.05° spatial resolution (to better capture geomorphological features; Supplement S4) and per decade, have numerous potential applications in biogeography, ecology, evolutionary biology and climate change research, which can enhance our understanding of anthropogenic impacts on biodiversity and support well-informed conservation and management strategies. In particular, the layers can be used in Species Distribution Modelling (SDM) (Peterson et al., 2011) to predict the distribution of species at the global scale (Chefaoui et al., 2015; Fragkopoulou et al., 2022), including non-native species (Assis et al., 2015), address niche-based questions (Hu et al., 2021; Lee-Yaw et al., 2016; Song et al., 2021) and phylogeographic hypotheses (Neiva et al., 2014), identify biodiversity hotspots (Fragkopoulou et al., 2022) and support the conservation and management of marine biodiversity (Boavida et al., 2016; Hobday et al., 2010). In the scope of SDM, by letting users filter historical data into two time periods (decades 2000–2010 and 2010–2020), the current version allows generating independent data for temporal cross-validation, which can assist in evaluating model performance and prediction error (Ko et al., 2013). Moreover, the development of biologically meaningful variables for future climate change scenarios (e.g. dissolved oxygen, primary productivity and pH) allows more realistic estimates of the anthropogenic pressures that may lead to extinction and turnover of populations (Assis, Serrão, et al., 2022; Gouvêa et al., 2022; Martins et al., 2021).

The marine layers can be used in additional analyses beyond SDM, such as those based on univariate algorithms of climate velocity

(Burrows et al., 2014) or multivariate algorithms of climate analogues (Mahony et al., 2017), to estimate biodiversity exposure to climate change (Mackintosh et al., 2023), or identify climate connectivity corridors promoting future distribution range shifts (García Molinos et al., 2016). In this scope, the availability of the Shared Socioeconomic Pathway scenarios, comprising the mitigation strategies of the Paris Agreement, coupled with the ensemble of multiple ESMs, allows supporting international climate policies and agendas (Martins et al., 2021). The increased interoperability via the ERDDAP data server (Wilson et al., 2020) opens the door for Bio-Oracle to be used in digital twinning, such as the European Digital Twin of the Ocean, or private initiatives like the Microsoft Planetary Computer. This aligns with the United Nations Decade of Ocean Science for Sustainable Development, which aims to boost the development of Digital Twins of the Ocean for supporting blue growth and global governance.

AUTHOR CONTRIBUTIONS

Jorge Assis, Lennert Tyberghein, Heroen Verbruggen and Olivier De Clerck conceived the study. Jorge Assis designed the data pipelines and wrote the manuscript. Vinícius W. Salazar, Frederic Leclercq and Bart Vanhoorne set up the ERDDAP server and Frederic Leclercq and Bart Vanhoorne will maintain it. Salvador Jesús Fernández Bejarano and Vinícius W. Salazar wrote the documentation, published the data set on ERDDAP and developed the R and Python clients. All authors reviewed the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declared no conflict of interest.

DATA AVAILABILITY STATEMENT

The layers are openly available at the ERDDAP instance hosted at the Flanders Marine Institute Data Centre (<https://erddap.bio-oracle.org/erddap>), through the Bio-ORACLE website (<https://www.bio-oracle.org>) and the R and Python packages available in the permanent GitHub repository (<https://github.com/bio-oracle>).

CODE AVAILABILITY

The R language code used to generate the layers is available in the permanent GitHub repository at <https://github.com/bio-oracle/data-layers>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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